

EXPERIMENT 1

Student Name: Hunar Katyal

UID: 24BDA70085

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment 1 – Understanding Networking Fundamentals using Cisco Packet Tracer

Aim

To understand the fundamentals of computer networking by studying IP addressing, Cisco IOS, Ethernet cables, RJ-45 connectors, Layer-2 switches, and routers, and to demonstrate their working using Cisco Packet Tracer.

Objective

- To understand the concept of IP addressing and subnet masks.
- To study the Cisco IOS Command Line Interface (CLI).
- To understand the difference between Straight-through and Cross-over cables.
- To identify the purpose of the RJ-45 connector.
- To understand the working of a Layer-2 Switch.
- To understand the role of a Router in connecting different networks.
- To verify network communication using the Ping command.

Software Requirements

Software

- Cisco Packet Tracer
- Windows Operating System

Hardware (Virtual)

- PC-PT
- Cisco 2960-24TT Switch
- Cisco 1941 Router

Practical / Experiment Steps

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Part I – IP Address

1. Place two PCs and one Layer-2 Switch in Cisco Packet Tracer.
2. Connect the PCs to the switch using Copper Straight-Through cables.
3. Assign IP addresses to both PCs.
4. Verify connectivity using the Ping command.

Part II – Cisco IOS

5. Add a Cisco 1941 Router.
6. Access the CLI.
7. Explore User EXEC Mode, Privileged EXEC Mode, and Global Configuration Mode.
8. Execute basic Cisco IOS commands.

Part III – Straight Cable & Cross Cable

9. Demonstrate Straight-through cable connections.
10. Demonstrate Cross-over cable connections.
11. Observe the difference between both cable types.

Part IV – RJ-45 Connector

12. Study the RJ-45 Ethernet connector and its purpose in networking.

Part V – Layer-2 Switch

13. Connect multiple PCs to a Layer-2 Switch.
14. Verify communication between all connected hosts.

Part VI – Router

15. Connect two different networks using a router.
16. Configure router interfaces.
17. Verify communication between the two networks.

Procedure

IP Address

1. Open Cisco Packet Tracer.

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2. Add two PCs and one Switch.
3. Connect using Copper Straight-Through cable.
4. Configure:

PC0

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0

PC1

- IP Address: 192.168.1.20
- Subnet Mask: 255.255.255.0

5. Open Command Prompt.

6. Execute:

ping 192.168.1.20

Cisco IOS

1. Add a Cisco 1941 Router.
2. Open CLI.
3. Type:

enable

4. Execute:

show running-config

show startup-config

show ip interface brief

5. Enter Global Configuration Mode.

configure terminal

6. Change hostname.

hostname R1

7. Save configuration.

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copy running-config startup-config

Straight Cable

1. Connect PC → Switch using Copper Straight-Through cable.
2. Connect Switch → Router using Copper Straight-Through cable.

Cross Cable

1. Connect PC → PC using Copper Cross-Over cable.
2. Connect Switch → Switch using Copper Cross-Over cable.

Layer-2 Switch

1. Connect three PCs to a Layer-2 Switch.
2. Assign IP addresses.
3. Verify communication using Ping.

Router

1. Connect two switches to a router.
2. Configure router interfaces.
3. Configure default gateways on PCs.
4. Verify communication using Ping.

Input / Output Details

Devices Used

- PC-PT
- Cisco 2960-24TT Switch
- Cisco 1941 Router

Cable Types

- Copper Straight-Through
- Copper Cross-Over

Cisco IOS Commands Used

enable

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configure terminal

hostname R1

show running-config

show startup-config

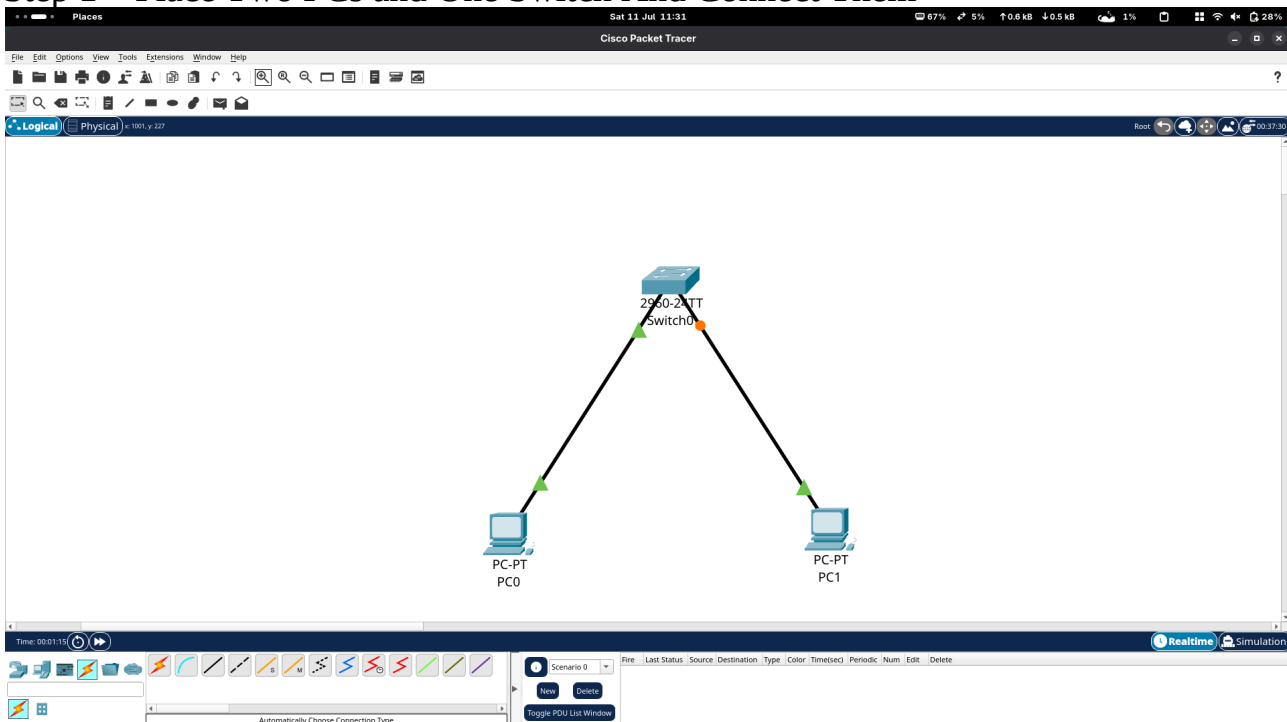
show ip interface brief

copy running-config startup-config

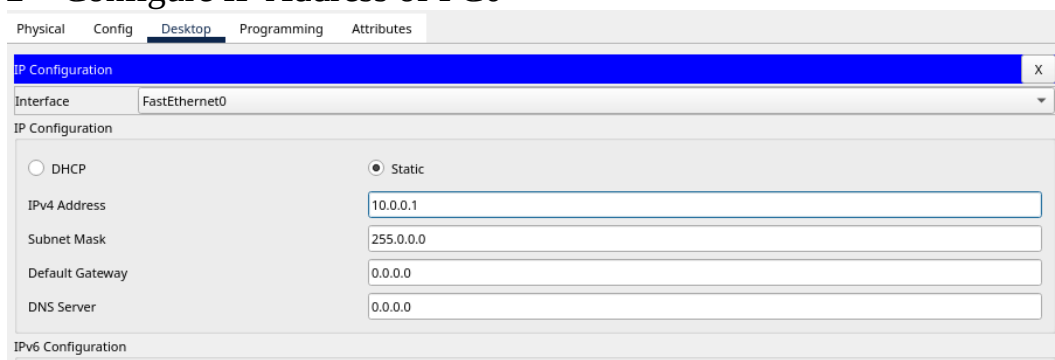
ping

Step-wise Output

Step 1 – Place Two PCs and One Switch And Connect Them



Step 2 – Configure IP Address of PC0



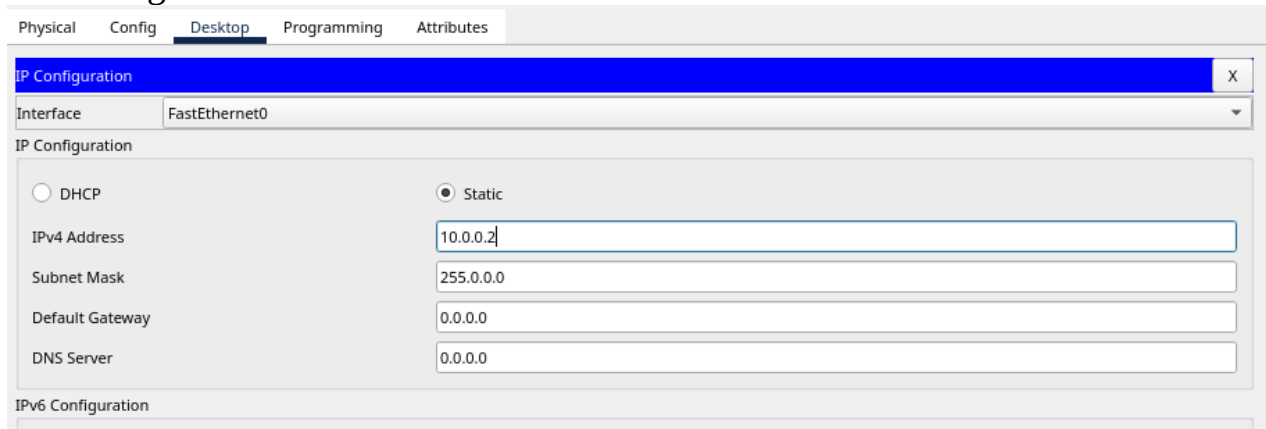
The screenshot shows the 'IP Configuration' window for PC0. The 'Interface' is 'FastEthernet0'. Under 'IP Configuration', the 'Static' option is selected. The fields are filled with the following values:

Field	Value
IPv4 Address	10.0.0.1
Subnet Mask	255.0.0.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

The 'IPv6 Configuration' section is visible at the bottom but is not filled out.

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Step 3 – Configure IP Address of PC1



Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 10.0.0.2

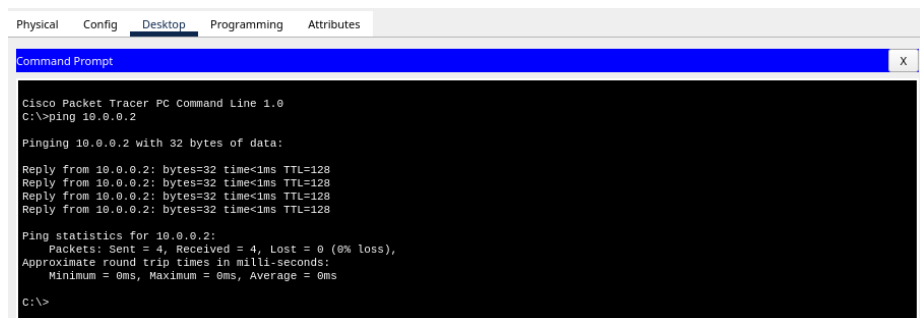
Subnet Mask: 255.0.0.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

Step 5 – Verify Communication Using Ping



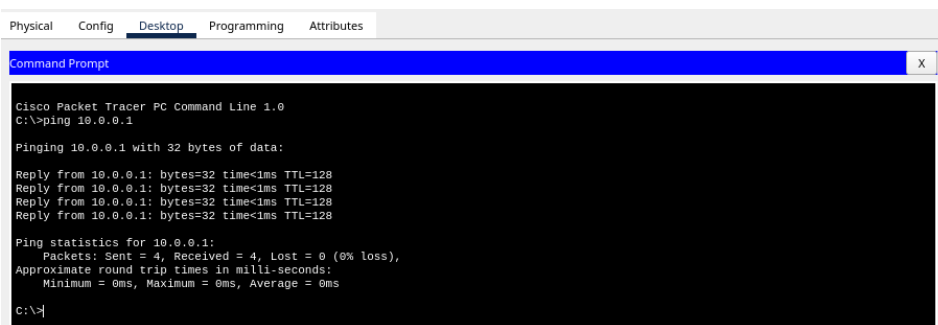
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

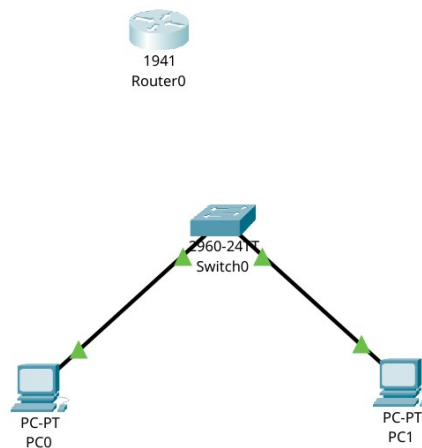
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Step 6 – Add Cisco 1941 Router

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Step 7 – User EXEC Mode (Router>)

```

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A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
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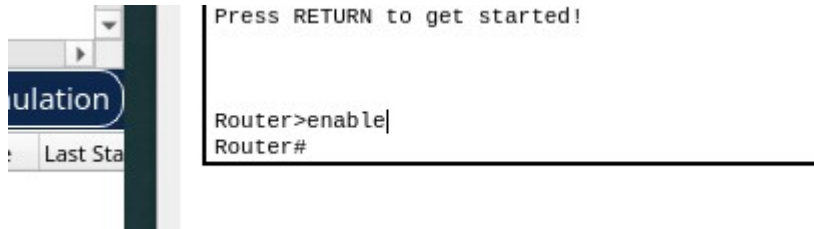
Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no
  
```

Step 8 – Privileged EXEC Mode (Router#)

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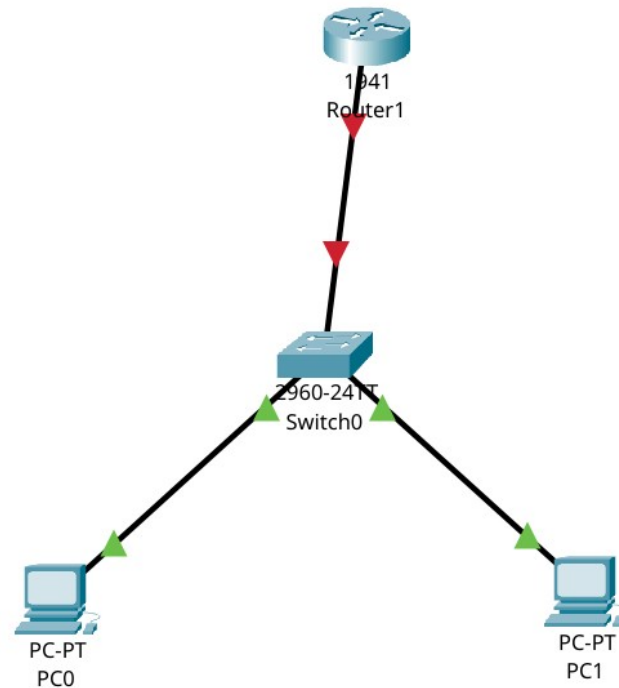


Step 9 – Global Configuration Mode (Router(config)#)

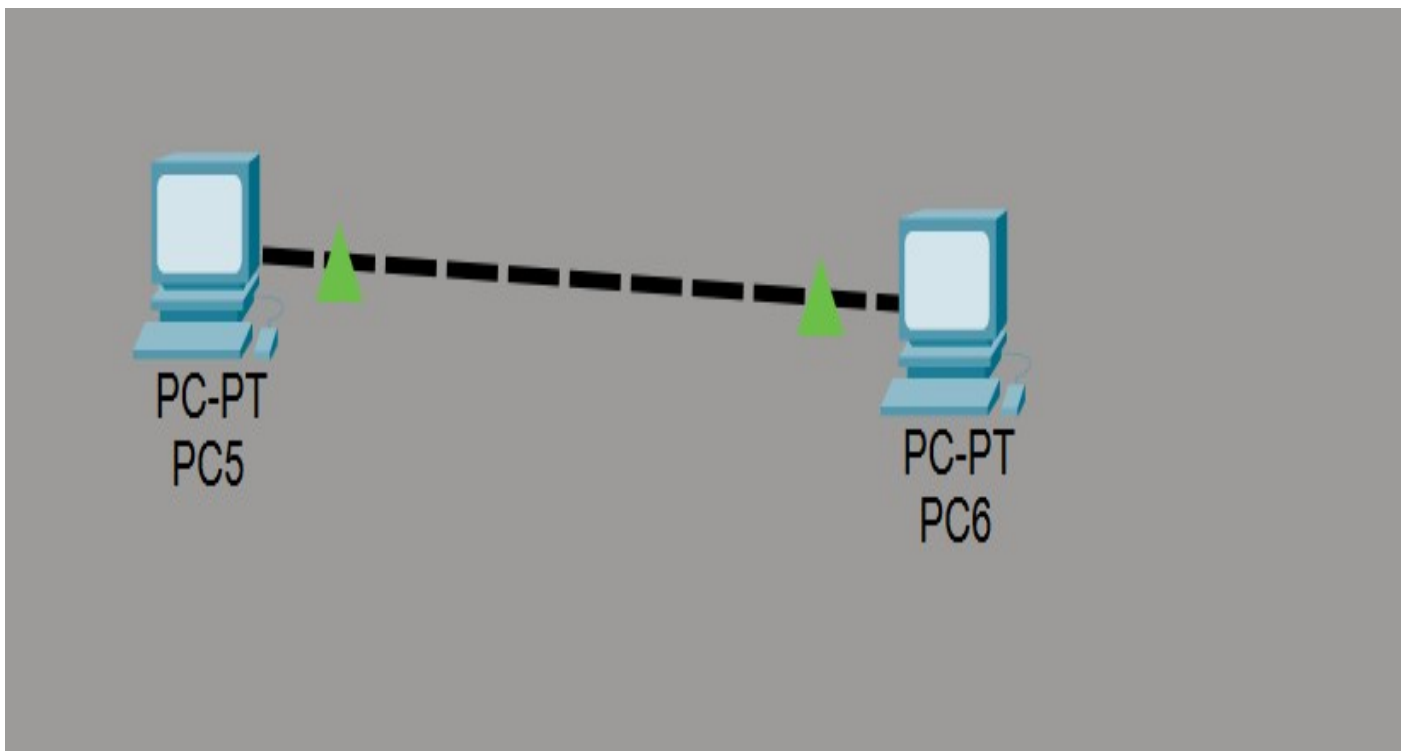
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname r1
r1(config)#enable secret 123
r1(config)#
```

Step 11 – Straight Cable Demonstration

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Step 12 – Cross Cable Demonstration

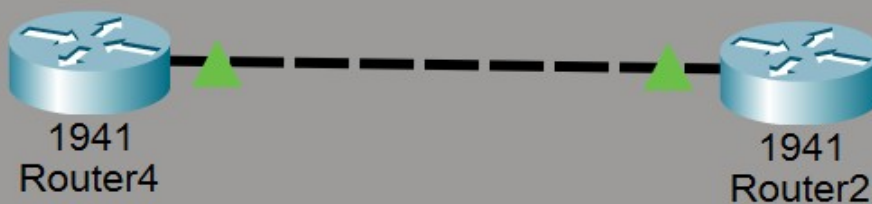


Step 13 – Layer-2 Switch Demonstration

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Step 14 – Router Demonstration



Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the concept of IP addressing and subnet masks.
- Configure basic network topologies in Cisco Packet Tracer.
- Use Cisco IOS CLI commands for router configuration.

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- Differentiate between Straight-through and Cross-over cables.
- Understand the purpose of the RJ-45 connector.
- Explain the working of a Layer-2 Switch.
- Configure a Router to connect different networks.
- Verify successful communication using the Ping command.
- Develop a basic understanding of computer network configuration and troubleshooting.